SE 811: Software Maintenance

Toukir Ahammed

Legacy Information Systems

Legacy Information Systems

- Legacy software systems are "large software systems that we don't know how to cope with but that are vital to our organization."
- This is the Phaseout stage of the *staged model.*
- A legacy system as "any information system that significantly resists modification and evolution to meet new and constantly changing business requirements."
- In addition, these information systems are mission-critical—that is, essential to the organization's business—and must be operational at all times.
- In summary, due to the lack of available skills and flexibility of the system, there is a strong urge to get rid of the legacy software system as soon as possible and start with something modern.

Legacy Information Systems

- A set of acceptable features of a legacy system:
  - large with millions of lines of code;
  - geriatric, often more than 10 years old;
  - written in obsolete programming languages;
  - lack of consistent documentation;
  - poor management of data, often based on flat-file structures;
  - degraded structure following years of modifications;
  - very difficult, if not impossible, to expand;
  - runs on old processors which are generally much slower; and
  - the support team is totally reliant on the expertise and knowledge of expert individuals.

Solutions For Legacy Information System (LIS)

- *Freeze.*
- *Outsource.*
- *Carry on maintenance.*
- *Discard and redevelop.*
- *Wrap.*
- *Migrate.*

Migration

- In general, migration comprises three main steps:
  (i) conversion of the existing schema to a target schema;
  (ii) conversion of data; and
  (iii) conversion of program.
- Schema conversion means translating the legacy database schema into an equivalent database structure expressed in the new technology.
- Data conversion means movement of the data instances from the legacy database to the target database.
- Data conversion requires three steps: *extract*, *transform*, and *load* (ETL)
- Program conversion means modifying a program to access the migrated database instead of the legacy data.

Migration Planning

- Step 1: Perform portfolio analysis.
- Step 2: Identify the stakeholders.
- Step 3: Understand the requirements.
- Step 4: Create a business case.
- Step 5: Make a go or no-go decision.
- Step 6: Understand the LIS.
- Step 7: Understand the target technology.
- Step 8: Evaluate the available technologies.
- Step 9: Define the target architecture.
- Step 10: Define a strategy.
- Step 11: Reconcile the strategy with the needs of the stakeholder.
- Step 12: Determine the resources required.

Perform portfolio analysis

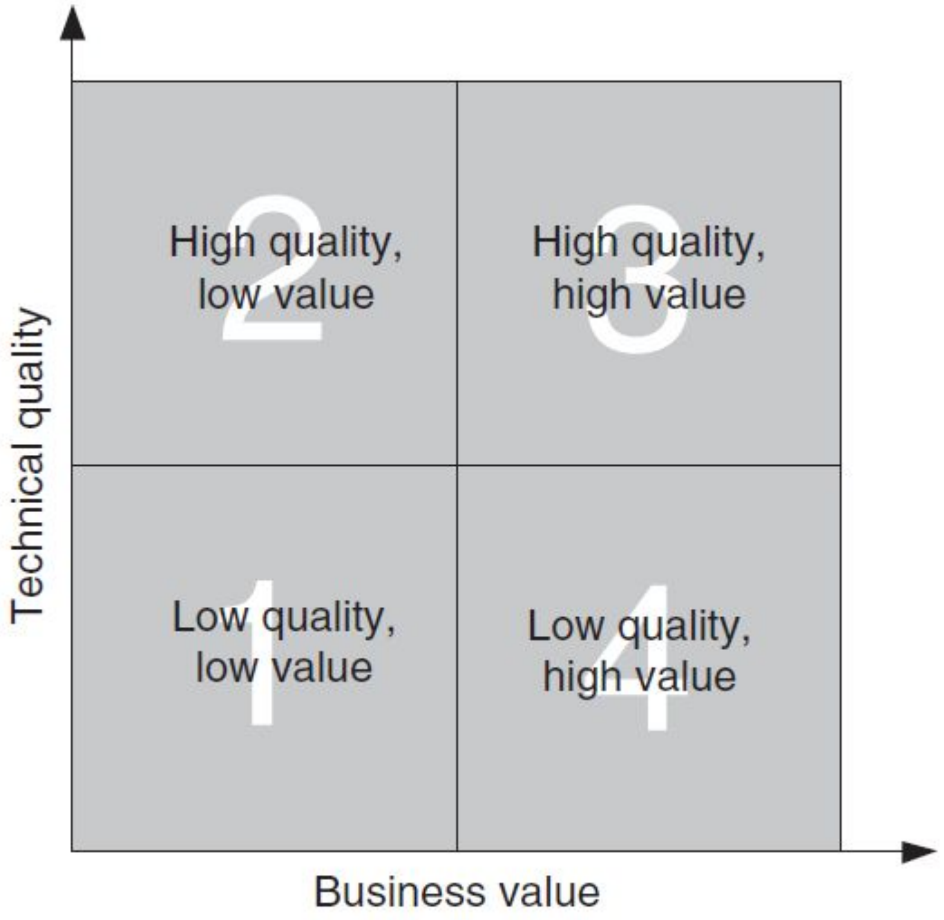

FIGURE 5.5 Portfolio analysis chi-square chart

- *Quadrant 1.* prime candidate for substitution with a commercial product if one is available.
- *Quadrant 2.* need not be replaced, migrated, or restructured.
- *Quadrant 3.* should be actively evolved.
- *Quadrant 4.* good candidate for redevelopment or migration to a new system.

Perform portfolio analysis

- criteria to establish business value:
  - level of usage, number of business goal satisfied, contribution to profit, user satisfaction, and annual revenue.
- criteria for technical quality:
  - ease of making changes, frequency of release notes, accuracy, performance of the system, error rate, cyclomatic complexity, and availability of training
- For example, if 100 is the maximum value of a certain criteria, and an application scores 30, then the coefficient of the said criteria is 0.30, which is obtained by dividing 30 by 100.

$$r = 1 - \left(\frac{actual\ measure - lower\_bound}{upper\_bound - lower\_bound} \right)$$